

MTU_Value*Service* **Technical Documentation**

Diesel Engine
12 V 595 TE70, TE 70L, TE 90
16 V 595 TE70, TE70L, TE90

Functional Description
M013024/01E



Printed in Germany

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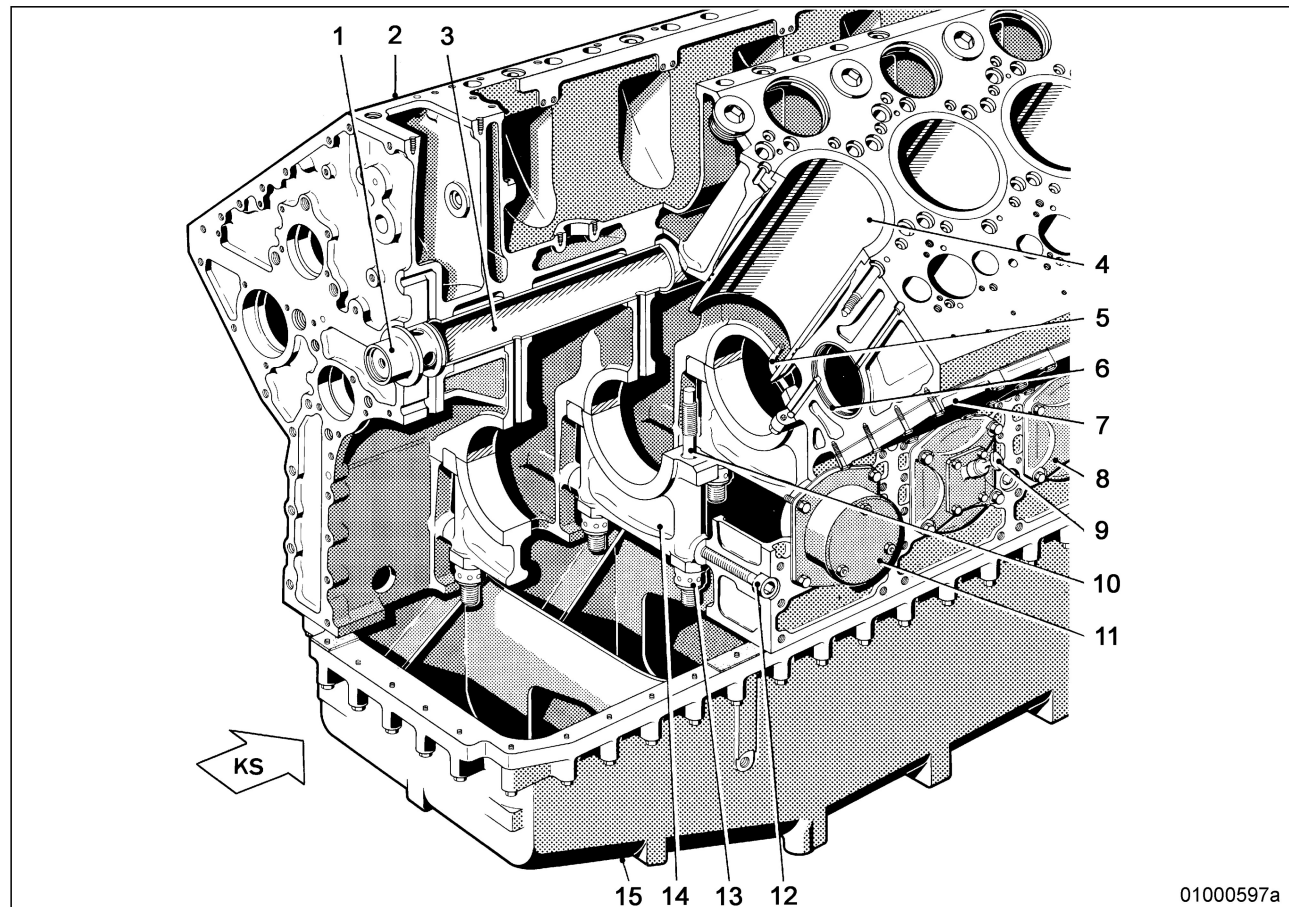
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Subject to alterations and amendments.

1	Series 595 engines	02
1.1	Crankcase	02
1.2	Gear train	04
1.3	Crank drive	06
1.4	Cylinder head and valve gear	08
1.5	Engine mounting	12
1.6	Fuel system	14
1.7	Charge-air and exhaust system	18
1.8	Compressed air supply system	28
1.9	Starting system	30
1.10	Lube oil system	32
1.11	Cooling system	42
1.12	High-temperature circuit	46
1.13	Low-temperature circuit	48
1.14	Engine coolant preheating	52
1.15	Raw-water system	56
1.16	Engine management and monitoring	58
1.17	Index	61

1 Series 595 engines

1.1 Crankcase



- 1 Pressure relief valve
- 2 Crankcase
- 3 Main oil gallery
- 4 Cylinder liner
- 5 Spray nozzle
- 6 Camshaft bearing

- 7 Cover
- 8 Inspection port cover
- 9 Oil dipstick
- 10 Main bearing cap stud
- 11 Safety valve
- 12 Cross tie bolt

- 13 Main bearing cap nut
- 14 Crankshaft bearing cap
- 15 Oil pan
- KS Driving end

Crankcase

All main engine components are mounted on the crankcase.

The crank chamber is sealed at the top by the crankcase lower deck in which the cylinder liner lower fits are machined.

Technical data

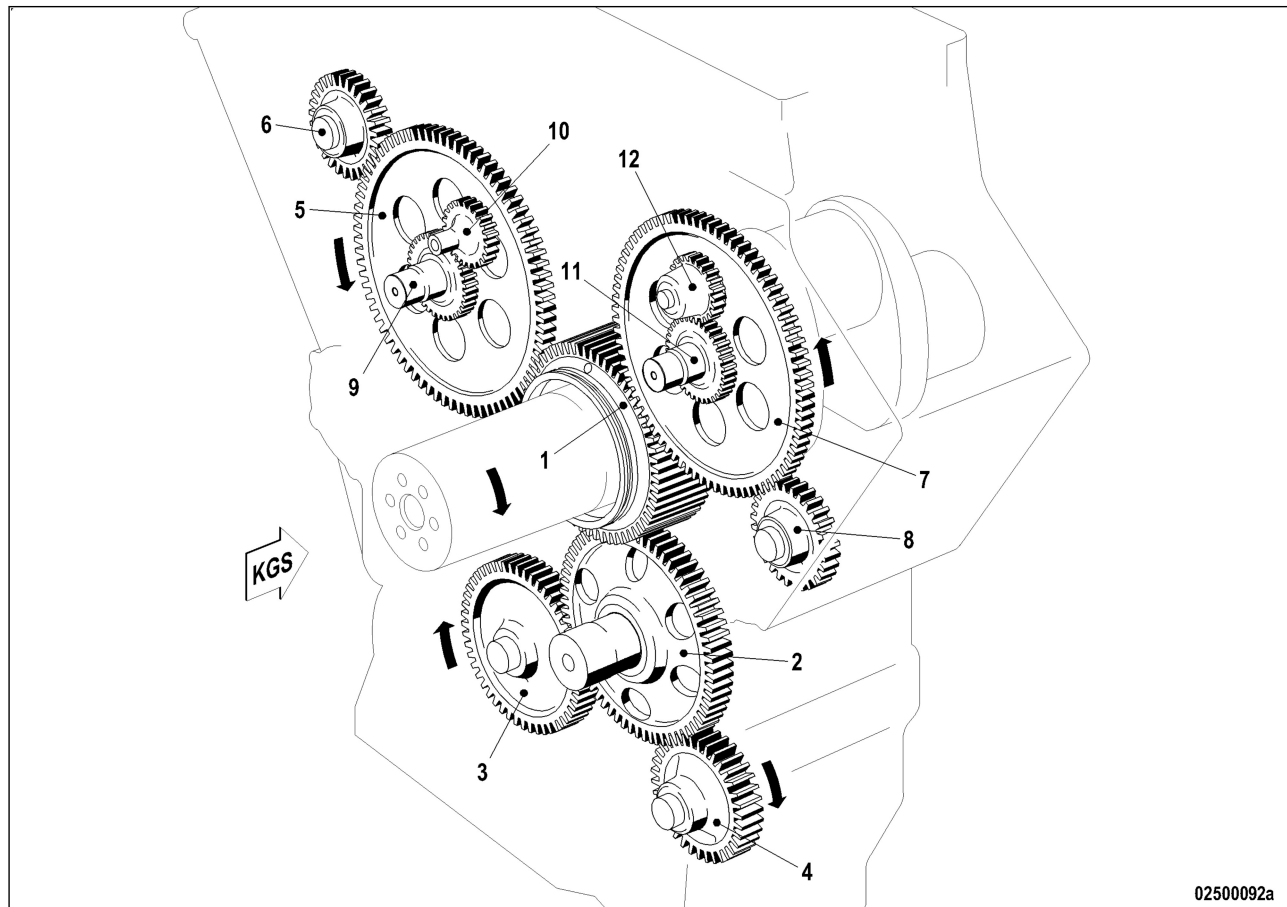
- Side walls extend below middle of crankshaft
- Swaged in steel tubes as main lube oil gallery
- Split sleeve bearings for crankshaft
- Sleeve bearings for camshaft
- Inspection port covers, partly with safety valve

Benefits

- High rigidity

-
- Low noise and vibration levels
 - Good accessibility for inspection and maintenance work

Gear train, free end



- | | | |
|------------------------------|---|----------------------------------|
| 1 Crankshaft gear | 6 Engine coolant pump drive gear | 11 Small countershaft gear |
| 2 Idler gear | 7 Large countershaft gear | 12 Fuel delivery pump drive gear |
| 3 Engine oil pump drive gear | 8 Charge-air coolant pump drive gear | Free end Free end |
| 4 Raw-water pump drive gear | 9 Small countershaft gear | |
| 5 Large countershaft gear | 10 Oil-level regulating pump drive gear | |

Gear train

The following components are driven by the gear train (driving end):

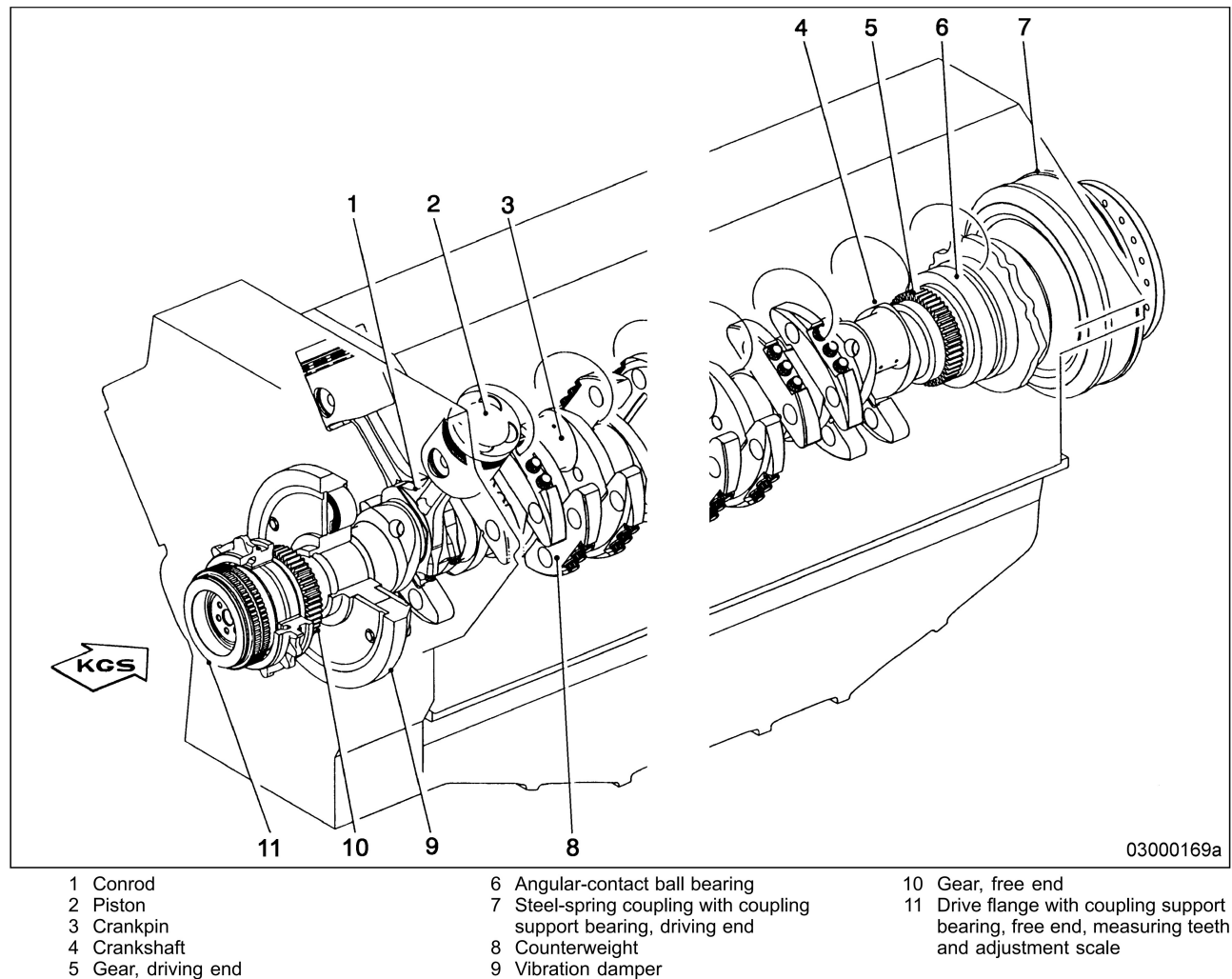
- Governor oil pump
- Compressed-air distributor
- Camshafts

The following components are driven by the gear train (free end):

- Oil pumps
- Coolant pumps
- Raw-water pump
- Fuel delivery pump

The gear train is located in the gearcase on the driving end and free end.

1.3 Crank drive



Crank drive

The forces generated in the combustion chambers of the cylinders are transmitted via pistons (2) and conrods (1) to the crankshaft (4), converted into rotary motion and output via the PTO flange. The components are carefully matched to one another and ensure high power with minimum wear.

Technical data

Piston

- Light-metal piston skirt
- Steel piston crown, secured to the skirt with stress bolts
- Three compression rings, one oil-scraper ring
- Piston cooling via oil spray nozzles

Conrod

- Forged
- Split bearing shells
- Oil supply to bearing via crankshaft

Crankshaft

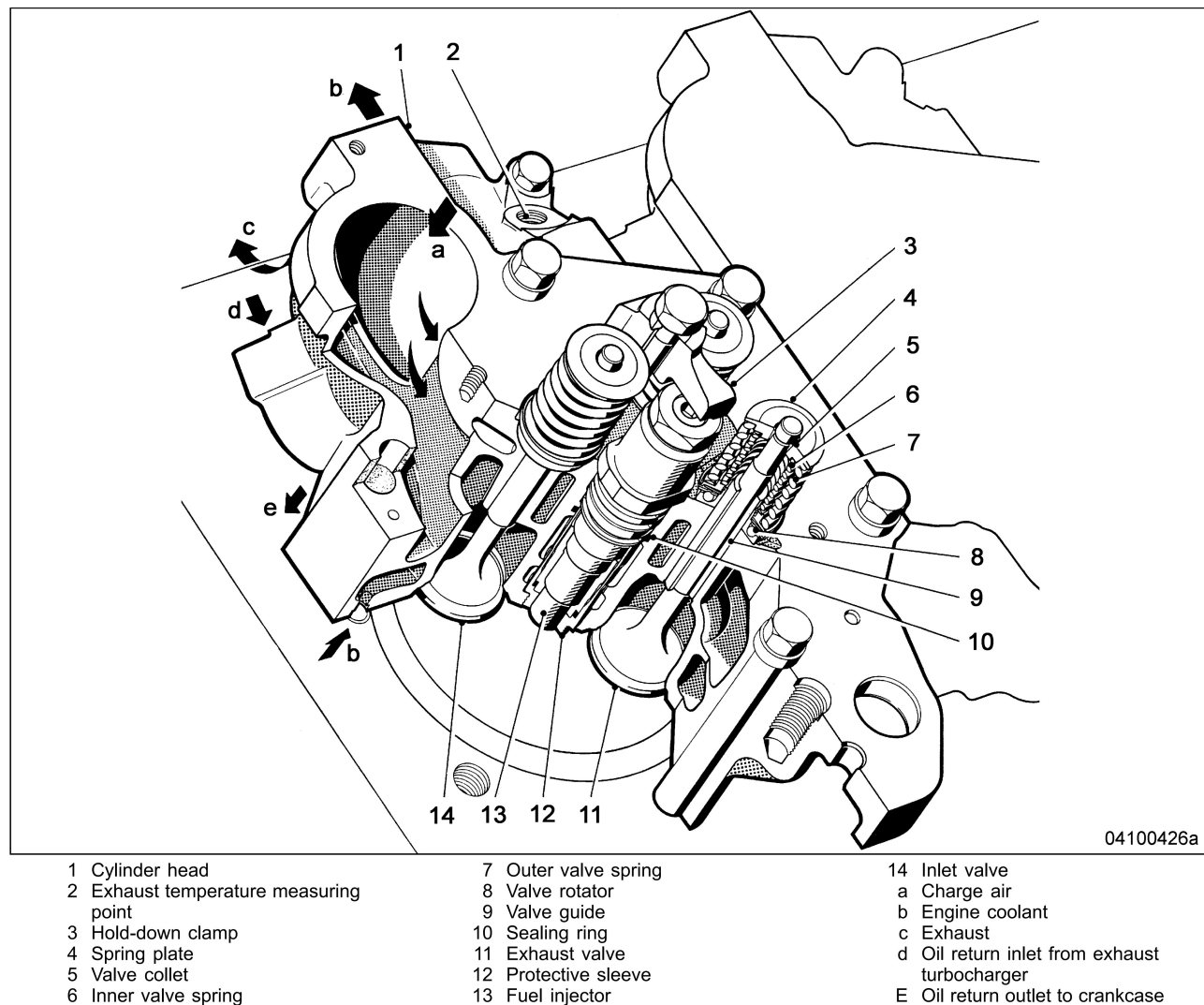
- Single-piece forging
- Bolt-on counterweights
- Mounted steel-spring coupling
- Press-fitted crankshaft gears
- Press-fitted drive flange on free end for power output
- Oil supply from lube oil system

Benefits

- High performance
- Long maintenance intervals
- Long service life

1.4 Cylinder head and valve gear

Cylinder head



Cylinder head

The cylinder heads (1) are secured to the crankcase and seal off the tops of the combustion chambers.

The joint between crankcase and cylinder head (1) is sealed with a gasket.

Engine coolant is supplied from the crankcase to cool the cylinder heads.

Fuel is supplied from the injection pumps to the injectors through the HP fuel lines.

Technology

- Individual cylinder heads
- Two inlet and two exhaust valves per cylinder
- Exhaust valves and inlet valves with valve rotator
- Centrally arranged injectors
- Transitions between crankcase and cylinder head sealed with gasket